

ARFAT NAIK

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PROFESSIONAL SUMMARY

Results-driven Artificial Intelligence and Machine Learning student with hands-on experience building, training, and deploying end-to-end ML systems on real-world datasets, alongside practical experience in local LLM inference and optimization. Proficient in Python, Scikit-learn, XGBoost, TensorFlow, Pandas, Flask, Docker, and MLflow. Experienced in supervised learning, feature engineering, NLP-based recommendation systems, automated ML pipelines, experiment tracking, and containerized deployments on cloud platforms.

TECHNICAL SKILLS

Languages: Python (NumPy, Pandas, Scikit-learn, XGBoost, TensorFlow, Joblib), SQL, JavaScript
Machine Learning: Supervised Learning (Regression, Classification), Feature Engineering, Model Training & Evaluation, Cross-Validation, Hyperparameter Tuning, Class Imbalance Handling
Performance Metrics: ROC-AUC, Precision, Recall, F1-Score, R^2 , RMSE, MAE
NLP & Recommendation: TF-IDF Vectorization, Cosine Similarity, Content-Based Filtering, Sparse Matrix Optimization
Data Science: EDA, Data Visualization (Matplotlib, Seaborn, Chart.js), Statistical Analysis, Data Cleaning, Outlier Detection, Missing Value Imputation
Local LLM Inference: llama.cpp, GGUF/IQ Quantization, MoE Architectures, KV-Cache & VRAM Optimization
MLOps & Tools: Flask (Model Serving), MLflow (Experiment Tracking), GitHub Actions (CI/CD), Redis, PostgreSQL (Supabase), Docker, Git, Jupyter Notebook, VS Code

PROJECTS

- AtmosIQ - AQI Forecasting** | *Python, XGBoost, MLflow, Flask, Redis, Docker, GitHub Actions* Live | GitHub
– Achieved $R^2 = 0.85$ and $MAE = 13.37$ on 3-hour AQI forecasting by training an **XGBoost regression model** with engineered lag features, rolling window statistics, and meteorological variables, selecting the best configuration across **22 MLflow-tracked experiments** (XGBoost and Random Forest variants).
– Eliminated all manual intervention in the ML lifecycle by building a **fully automated hourly inference pipeline** with GitHub Actions, using **Redis as a 3-hour rolling feature buffer** and **Supabase** for persistent storage — enabling continuous data ingestion, feature transformation, prediction, and logging end-to-end.
– Reduced dashboard API response latency by implementing **Redis-backed caching** (5-min TTL) shared across Gunicorn workers, and deployed **8 REST endpoints** via a **Dockerized Flask app on Railway** with Chart.js visualizations for real-time PM2.5, PM10, and forecast insights.
- Credit Risk Prediction System** | *Python, Logistic Regression, Scikit-learn, Flask, Docker* Live | GitHub
– Maximized loan default detection recall by designing a **Logistic Regression model with cost-sensitive thresholding**, directly minimizing false approvals on a heavily imbalanced credit dataset.
– Ensured unbiased and production-reliable model evaluation by applying rigorous **data preprocessing**, missing value imputation, outlier handling, feature engineering, strict train-test leakage prevention, and **class imbalance management**, enabling robust generalization to unseen data.
– Delivered real-time default probability scoring to end users by containerizing a **Flask REST API** with **Docker** and deploying on **Railway**, serving binary credit decisions and probability scores through a live web interface.
- Movie Recommendation System** | *Python, TF-IDF, Cosine Similarity, Flask, Docker, NLP* Live | GitHub
– Reduced similarity computation runtime by **60%** for a **content-based NLP recommendation engine** across 5,000+ TMDB movies by replacing dense matrix operations with **sparse matrix representations** on TF-IDF vectors built from a multi-field feature corpus (genres, keywords, cast, director, overview).
– Enabled production-ready real-time recommendations by containerizing the **Flask API** with **Docker** and deploying on **Railway**, with input validation and graceful error handling ensuring reliability at the API layer.

EDUCATION

Rizvi College of Engineering Mumbai, India
B.E. in Electronics and Computer Engineering; Minor in Artificial Intelligence and Machine Learning 2023 – 2027
– CGPA: 7.51/10 | Semester 6
B.E. Minors: Artificial Intelligence and Machine Learning 2025 – 2027